Yusuf **Dalva**

Research & Teaching Assistant · Ph.D. Student

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Education

Ph.D. in Computer Science, Virginia Tech, Blacksburg, VA, United States

Aug. 2023 - June 2027 (Expected)

- Research focus: Controllability in diffusion models
- Under the supervision of Pinar Yanardag
- Related coursework: Embodied AI, Learning-based Computer Vision

M.S. in Computer Engineering, Bilkent University

Sep. 2020 - June 2023

- Thesis topic: Image-to-image Translation for Face Attribute Editing with Disentangled Latent Directions
- Under the supervision of Aysegul Dundar
- **Best Master Thesis Award** by IEEE Computer Society, Turkey Chapter
- Related coursework: Computer Vision, Deep Learning, Deep Generative Networks, Computer Graphics (CGPA: 4.00/4.00)
- Awarded **Department Scholarship** at the time of enrollment

B.Sc. in Computer Engineering, Bilkent University

Sep. 2016 - June 2020

- Graduation Project: DRIVision - Mobile-based Driving Assistance Solutions (**Data Science Award**)
- Related coursework: Object-Oriented Software Engineering, Algorithms, Operating Systems, Database Systems (CGPA: 3.67/4.00)
- Awarded **Merit Scholarship** in 2017, 2018, 2019

Publications

- **Y. Dalva** and P. Yanardag, *Adastate: Self-evolving anchors for streaming video generation*, 2026. arXiv: `2605.30349 [cs.CV]`. [Online]. Available: `https://arxiv.org/abs/2605.30349`
- **Y. Dalva**, H. Yesiltepe, and P. Yanardag, "Learn once, edit anywhere: Visual direction transfer for diffusion models," in *European Conference on Computer Vision*, 2026
- **Y.Dalva**, G. G. Qian, M. Goldenberg, *et al.*, "Canvas-to-image: Compositional image generation with multimodal controls," in *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers (SIGGRAPH)*, 2026
- **Y. Dalva**, H. Yesiltepe, and P. Yanardag, "LoRAShop: Training-free multi-concept image generation and editing with rectified flow transformers," in *Advances in Neural Information Processing Systems* ***(Spotlight - top 3%)***, 2025
- **Y. Dalva**, K. Venkatesh, and P. Yanardag, "Fluxspace: Disentangled semantic editing in rectified flow models," in *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)*, 2025
- **Y. Dalva** and P. Yanardag, "NoiseCLR: A contrastive learning approach for unsupervised discovery of interpretable directions in diffusion models," in *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)* ***(Oral - top 0.7%)***, 2024
- **Y. Dalva**, Y. Li, Q. Liu, *et al.*, "Layerfusion: Harmonized multi-layer text-to-image generation with generative priors," in *IEEE / CVF Computer Vision and Pattern Recognition Conference Findings (CVPRF)*, 2026
- K. Venkatesh, **Y. Dalva**, I. Lourentzou, *et al.*, "Context canvas: Enhancing text-to-image diffusion models with knowledge graph-based rag," *arXiv preprint arXiv:2412.09614*, 2024
- H. Yesiltepe, **Y. Dalva**, and P. Yanardag, "The curious case of end token: A zero-shot disentangled image editing using clip," *arXiv preprint arXiv:2406.00457*, 2024
- **Y. Dalva**, H. Pehlivan, O. I. Hatipoglu, *et al.*, "Image-to-Image Translation with Disentangled Latent Vectors for Face Editing," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023
- **Y. Dalva**, S. F. Altındiş, H. Pehlivan, *et al.*, "Benchmarking the Robustness of Instance Segmentation Models," in *IEEE Transactions on Neural Networks and Learning Systems*, 2023
- **Y. Dalva**, S. F. Altındiş, and A. Dundar, "VecGAN: Image-to-Image Translation with Interpretable Latent Directions," in *European Conference on Computer Vision*, Springer, 2022, pp. 153–169
- H. Pehlivan, **Y. Dalva**, and A. Dundar, "StyleRes: Transforming the Residuals for Real Image Editing with StyleGAN," in *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)*, 2023

Experience

Research Intern, Google Research *May 2026 - Present*

- Mentors: Mauricio Delbracio, Arpit Bansal, Ashwini Pokle
- Working on **efficient image tokenization** for generative models, exploring **compact latent representations** that reduce token counts while preserving **reconstruction fidelity** and downstream **generation quality**.

Research Intern, Snap Inc. *May 2025 - Dec. 2025*

- Mentors: Gordon Qian, Or Patashnik, Daniel Cohen-Or, Kfir Aberman, Jackson Wang, Sergey Tulyakov
- Developed a multi-subject diffusion framework enabling **spatially controllable composition** from canvas inputs and efficient multi-concept conditioning, incorporating **vision-language representations** to enhance **semantic understanding in generation**

Research Scientist/Engineer Intern, Adobe Inc. *May 2024 - Aug. 2024*

- Mentors: Yijun Li, Qing Liu, Nanxuan Zhao, Jianming Zhang, Zhe Lin
- Developed an **inference time harmonization** approach for layered image generators. Introduced the concept of **attention-level blending** for layered images to achieve layered compositions with interactions between layers.

Teaching Assistant, Virginia Tech *Jan. 2024 - May 2025*

- Assisted Courses: **AI Tools for Software Engineering**, **Intermediate Software Design and Engineering**
- Assisted students on projects on **software engineering principles** and connecting them with **ChatGPT-related tools**.

Teaching Assistant, Bilkent University *Sep. 2020 - June 2023*

- Won **Outstanding Teaching Assistant** award **3** times (2021, 2022, 2023)
- Gave tutorials on **Google Colab** and **PyTorch**
- Assisted Courses: **Introduction to Machine Learning**, Operating Systems, Computer Organization, **Algorithms and Programming I**

Software Engineer Intern, Atlassian, Opsgenie *July 2019 - Sep. 2019*

- Engaged in projects as a part of the Business Operations team
- Translated legacy records to Atlassian database
- Developed **AWS Lambdas** for subscription actions

Achievements, Honors & Awards

2025	**Amazon Fellowship** Selected as one of two fellows, as a part of Amazon - Virginia Tech Initiative in Efficient and Robust Machine Learning
2023	**Best Master Thesis Award** Selected as the best master thesis by IEEE Computer Society Turkey Chapter
2021 - 2023	**Outstanding Teaching Assistant** Chosen as one of the three most successful teaching assistants in the Department of Computer Engineering of Bilkent University

Conferences & Presentations

ViDiT – ECCV Poster, Malmo, Sweden	Sep. 2026
Canvas-to-Image – SIGGRAPH, Los Angeles, CA	Aug. 2026
LayerFusion – CVPR Findings, Denver, CO	June 2026
LoraShop – NeurIPS Spotlight, San Diego, CA	Dec. 2025
FluxSpace – CVPR Poster, Nashville, TN	June 2025
NoiseCLR – CVPR Oral + Poster, Seattle, WA	June 2024
StyleRes – CVPR Poster, Vancouver, Canada	June 2023
VecGAN – ECCV Poster, Tel Aviv, Israel	Sep. 2022

Academic Service

Conference Reviewer	CVPR (2024, 2025, 2026), ECCV 2026, ICCV 2025[†], NeurIPS (2025, 2026), ICLR (2024, 2026), ICML 2026[‡], BMVC 2024
Journal Reviewer	IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI, 2025, 2026)
	IEEE Transactions on Neural Networks and Learning Systems (TNNLS, 2025)
	International Journal of Computer Vision (IJCV, 2024)
Recognition	[†]Outstanding Reviewer (ICCV 2025, CVPR 2026, ECCV 2026), [‡]Gold Reviewer (ICML 2026)